

## *Data Exploration and Visualization*

Main aspects using plots and summary statistics

## 1 DATA EXPLORATION

- Data Preprocessing and Manipulation
- Numerical Summaries
- Data Visualization
- Other Exploratory Methods

# DATA EXPLORATION

## IMPORTANT MESSAGE

Before trying anything, you should know your data!

# WHAT IS DATA EXPLORATION?

- It is the preliminary exploration of the data aimed at identifying their most relevant characteristics.
- Two of the key motivations may be:
  - Helping to select the right tool for preprocessing.
  - Exploiting humans' abilities to recognize patterns not captured by automatic tools.
- Related to Exploratory Data Analysis (EDA), created by statistician John Tukey. His seminal work is *Exploratory Data Analysis*.

# EXPLORATORY DATA ANALYSIS (EDA)

- *An approach of analyzing data to summarize their main characteristics without using a statistical model or having formulated a prior hypothesis.*
- EDA was promoted by John Tukey to encourage statisticians to explore data and, possibly, formulate hypothesis that could lead to new data collection and experiments.
- Nice online introduction in Chapter 1 of the NIST Engineering Statistics Handbook ▶ <https://www.itl.nist.gov/div898/handbook/eda/section1/eda11.htm>
- [http://en.wikipedia.org/wiki/Exploratory\\_data\\_analysis](http://en.wikipedia.org/wiki/Exploratory_data_analysis)

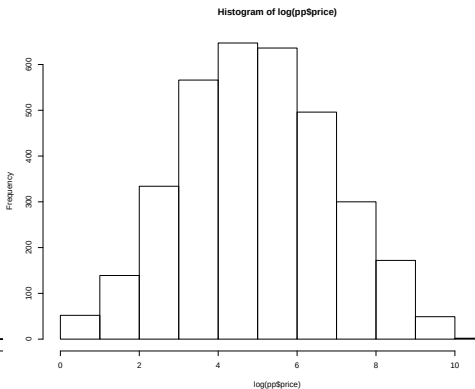
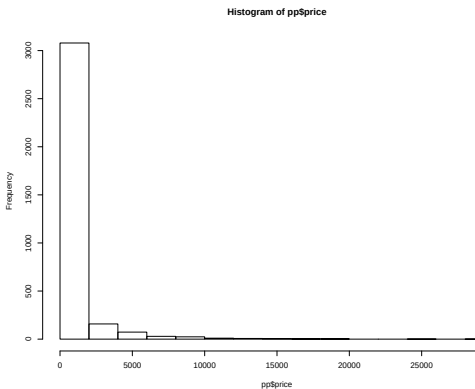
# DATA PREPROCESSING AND MANIPULATION

- Data cleaning: finding incomplete data, incorrect data, inconsistent data, duplicate data (major issue when merging data from heterogeneous sources). No quality in data, no quality in the results, (trash in, trash out).
- Data integration if you have to integrate multiple databases or files. Filtering or subsetting. Sampling.
- Aggregating: collapse multiple values into a single value, for instance, cities aggregated into regions, states or countries.

# DATA PREPROCESSING AND MANIPULATION

- Feature (variable) reduction or feature creation:
  - Redundant features (variables) that duplicate much or all of the information contained in one or more other attributes. For example: purchase price of a product and the amount of sales tax paid.
  - Irrelevant features that contain no useful information for the task at hand. For example: students' ID is often irrelevant to the task of predicting students' GPA.
  - Create new attributes that can capture the important information in a data set much more efficiently than the original attributes: for example, given the birthday, create the attribute age.
- Data transformation applying a mathematical operation to make your data more symmetric, more constant in variance, etc. Data discretization.

# TRANSFORMATIONS





# NUMERICAL SUMMARIES

- Numbers (statistics) that summarise properties of the data (variables), properties of their (univariate, multivariate or conditional) distributions.
- Summarized properties include location, spread, shape, dependence.
- Most summary statistics can be computed in a single pass but their interpretation is a little bit tricky.

*All our knowledge begins from sense (observation), passes understanding (description and modeling) and ends in reason (inference).*

Immanuel Kant

# DATA VISUALIZATION

Today, data visualization has become a rapidly evolving blend of science and art that is changing the corporate landscape:

- How to blend mathematics and cognitive science to produce effective visualizations.
- How to use visualizations to communicate research findings efficiently.

## IMPORTANT MESSAGE

An image is worth a thousand words.

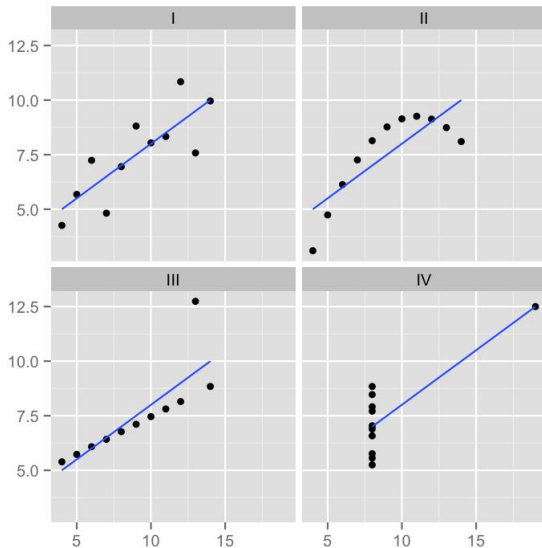
- Good visualizations draw attention to the most important point
- Unnecessary complexity distracts. Revise for simplicity.

- Visualization is the conversion of data into a visual or tabular format so that the characteristics of the data and the relationships between data items or attributes can be analyzed or reported.
- Data visualization is one of the most powerful and appealing techniques for data exploration.
  - Humans have a well developed ability to analyze large amounts of information that is presented visually.
  - It can detect general patterns and trends.
  - It can detect outliers and unusual patterns.
- Often, we get useful features including: skewness, multimodality, outliers or different groupings.

# CLASSICAL EXAMPLE ON THE IMPORTANCE OF VISUALIZATION

These four datasets emphasizes the importance of using graphs in (statistical) data analysis. All sets have 11 observations, the same mean of  $X(9)$  and  $Y(7,5)$ , the same fitted regression line ( $Y = 3 + 0,5X$ ), the same regression and residual sum of squares and therefore the same multiple R-squared of 0,67. But they represent very different situations.

# CLASSICAL EXAMPLE ON THE IMPORTANCE OF VISUALIZATION



# VISUALIZATION

It is a resource of the statistical method for data analysis. Statistical correction of the graphic information is key, choose suitable graphics for each type of data. If necessary, consult a [data visualization catalogue](#).

A graphic/ chart / plot should:

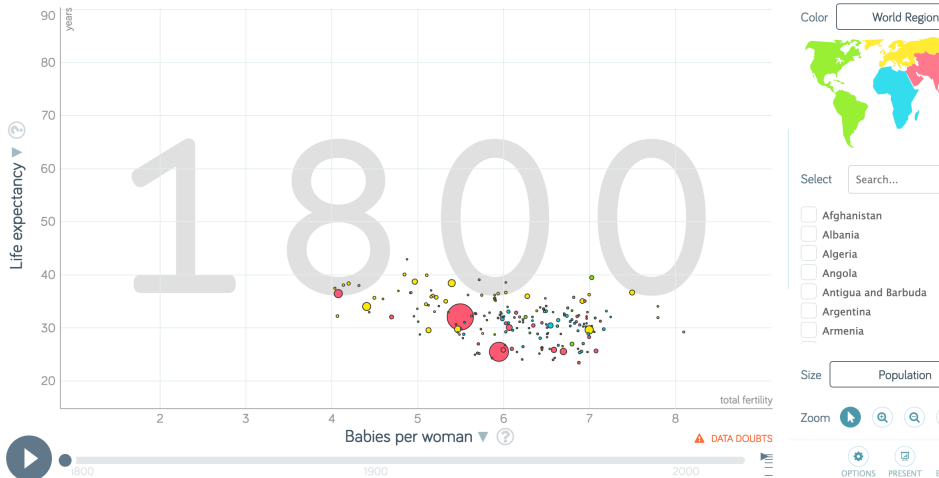
- Show the data and invite to think about the data.
- Do not alter the data (only changes of scale).
- Condense many data with few elements and reduced space.
- Invite to compare items that represent data properties.
- Reveal the nature of the data: from the general level to specific levels and details.
- Serve to describe, and explore the data and show reasonable aesthetic characteristics.
- Be integrated with the text that analyzes the data set.

# BASIC PLOTS

- The **boxplot**: a simple univariate device that detects outliers variable by variable and are great for comparing distributions of the same continuous variable across different groups.
- Two basic techniques for estimating densities: **histograms** and **kernel densities**. A density estimate gives a quick insight into the shape of the distribution of the data.
- **Scatterplots matrices** are very useful for plotting multiple variables against each other: they help to understand the nature of the relationship between variables in a data set and allow the detection of groups of clusters of points. Additional attributes can be displayed using the size, shape, color of the markers that represent the observations.
- **Parallel coordinate** plots are useful to detect outliers and/or groups.



# BASIC PLOTS - ENHANCED SCATTERPLOTS

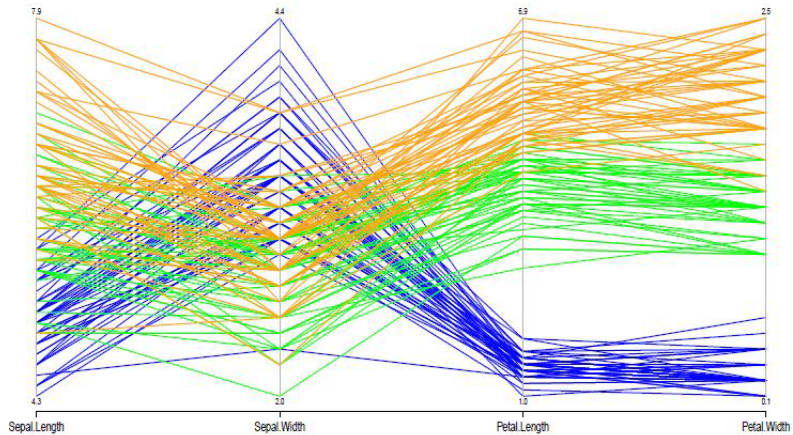


Watch Hans Rosling's videos (TED talks) on YouTube!

# PARALLEL COORDINATES PLOT

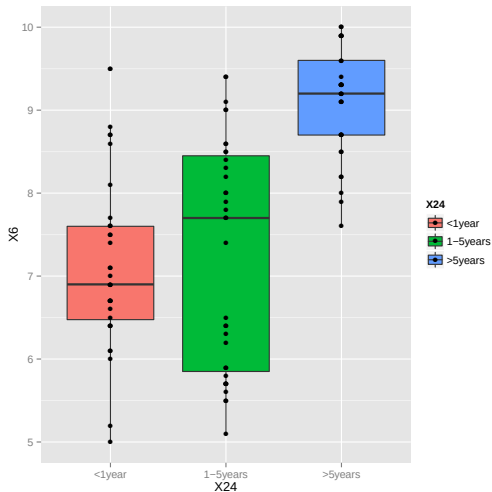
- Parallel Coordinates Plot (PCP) is a method for representing high-dimensional data.
- Instead of plotting observations in an orthogonal coordinate system, PCP draws coordinates in parallel axes and connects them with straight lines.
- The variables are drawn into the horizontal axis, and the values of the variables are mapped onto the vertical axis.
- The PCP is very useful for high dimensional data. However, it is sensitive to the order of the variables, since certain trends in the data can be shown more clearly in one ordering than in another.

# PARALLEL COORDINATE PLOT



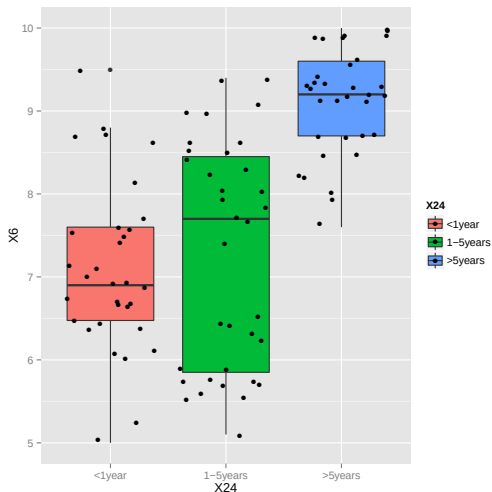
## OTHER TRICKS YOU CAN TRY

- Multiple boxplots: it will visually tell you if there are differences between groups in the distribution of a continuous variable. These plots are related to 1 way ANOVA (ANalysis Of VAriance) regression models.



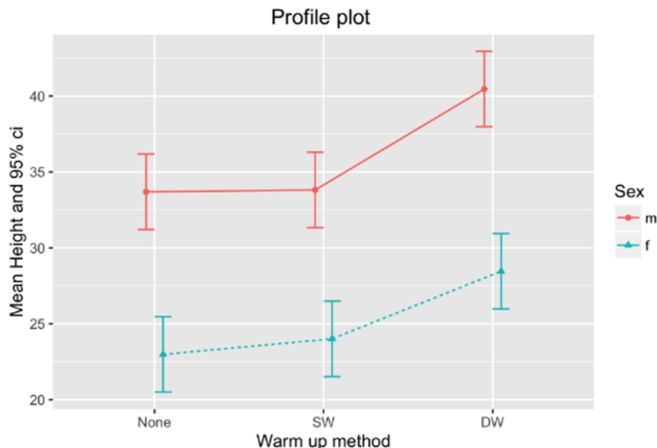
# OTHER TRICKS YOU CAN TRY

Using the *jitter* option.



# INTERACTION PLOTS

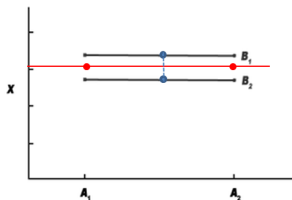
Or profile plots: they are related to 2 way ANOVA: one continuous response variable and two categorical variables.



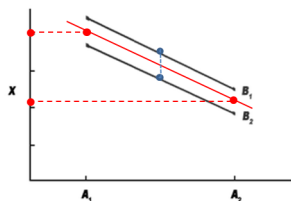
There is clear gender effect; the effect of warmup method is less clear; the lines are nearly parallel so there does not appear to be any evidence of interaction.

# INTERACTION PLOTS

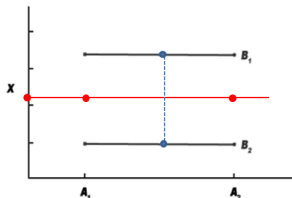
In the graph below there is no effect of Factor A, a small effect of Factor B (and if there were no effect of Factor B the two lines would coincide), and no interaction between Factor A and Factor B.



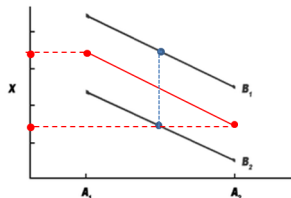
The graph below shows a large effect of Factor A, small effect of Factor B, and no interaction.



No effect of Factor A, larger effect of Factor B, and no interaction.



Here there is a large effect of Factor A, a large effect of Factor B and no interaction.



# INTERACTION PLOTS

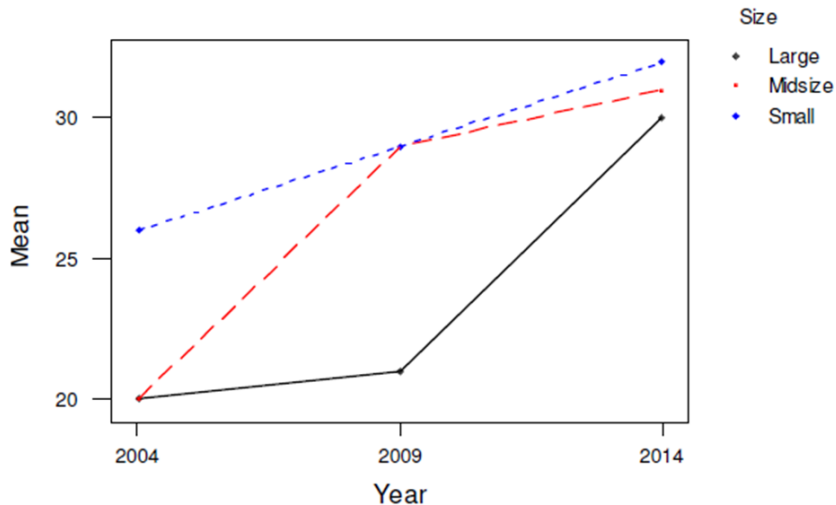
For instance, if we are interested in investigating the relationship between the mileage of new automobiles (in miles per gallon) and the year the auto was built (2004, 2009, or 2014) and the size of the auto (Small, Midsize, or Large).

| <i>Size</i>    | <i>2004</i>      |                         | <i>2009</i>      |                         | <i>2014</i>      | <i>Average</i> |
|----------------|------------------|-------------------------|------------------|-------------------------|------------------|----------------|
| Small          | 26               | $\longleftrightarrow^3$ | 29               | $\longleftrightarrow^3$ | 32               | 29             |
|                | $\updownarrow 6$ |                         | $\updownarrow 0$ |                         | $\updownarrow 1$ |                |
| Midsize        | 20               | $\longleftrightarrow^9$ | 29               | $\longleftrightarrow^2$ | 31               | 26.7           |
|                | $\updownarrow 0$ |                         | $\updownarrow 8$ |                         | $\updownarrow 1$ |                |
| Large          | 20               | $\longleftrightarrow^1$ | 21               | $\longleftrightarrow^9$ | 30               | 23.7           |
| <i>Average</i> | 22               |                         | 26.3             |                         | 31               | 26.4           |



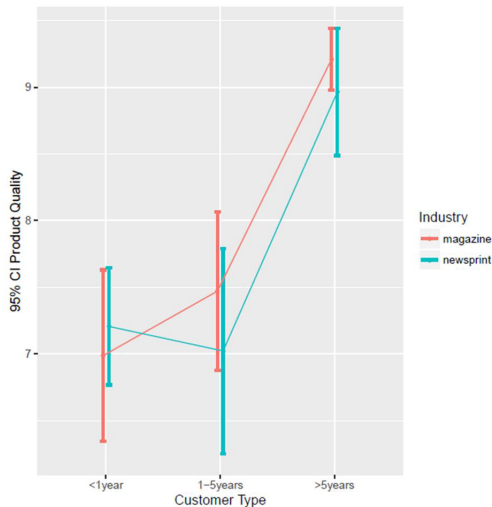
# INTERACTION PLOTS

Interaction Plot - Data Means for MPG



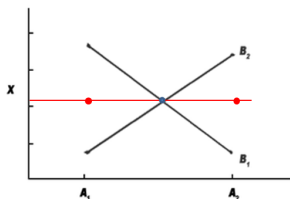
# INTERACTION PLOTS

Data set hbat: response variable is Product Quality Perception

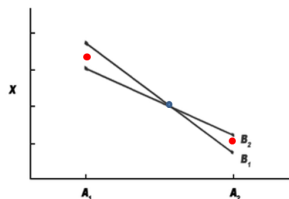


# INTERACTION PLOTS

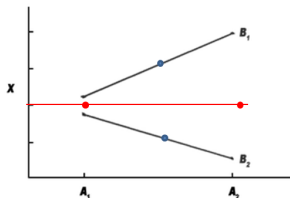
This is evidence of no effect of Factor A, no effect of Factor B but an interaction between A and B.



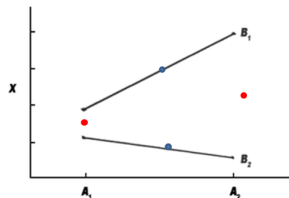
Large effect of Factor A, no effect of Factor B with a slight interaction.



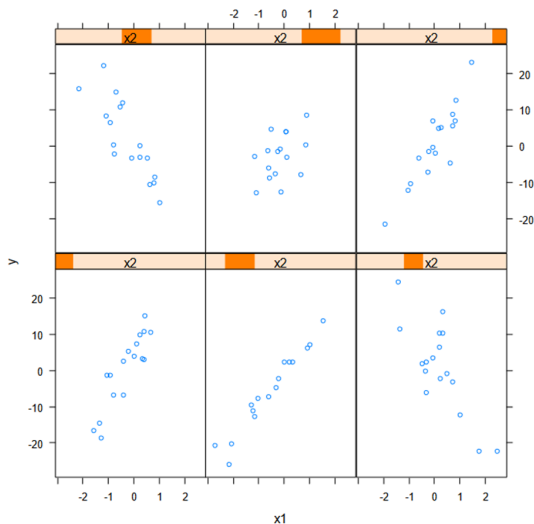
No effect of Factor A, a large effect of Factor B, with a very large interaction.



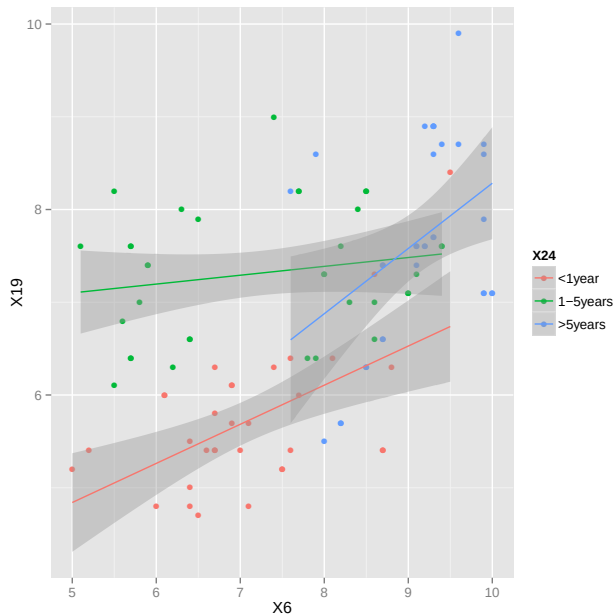
An effect of Factor A, a large effect of Factor B with a large interaction.



# CONDITIONING PLOTS (TO FIND INTERACTIONS)

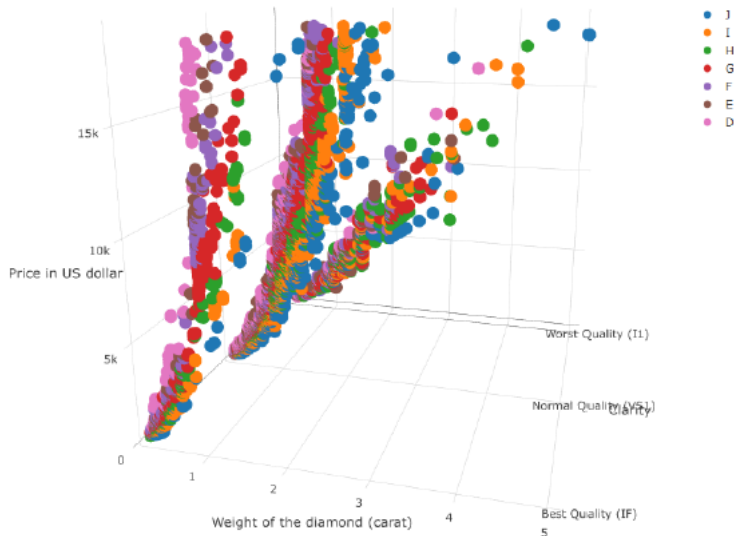


# DIFFERENT (LINEAR) MODELS



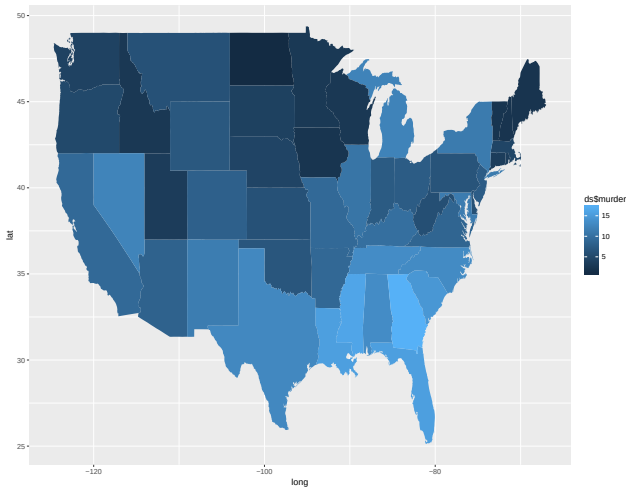
ANCOVA models

# 3D SCATTERPLOTS (LIBRARIES PLOT.LY, SHINY)

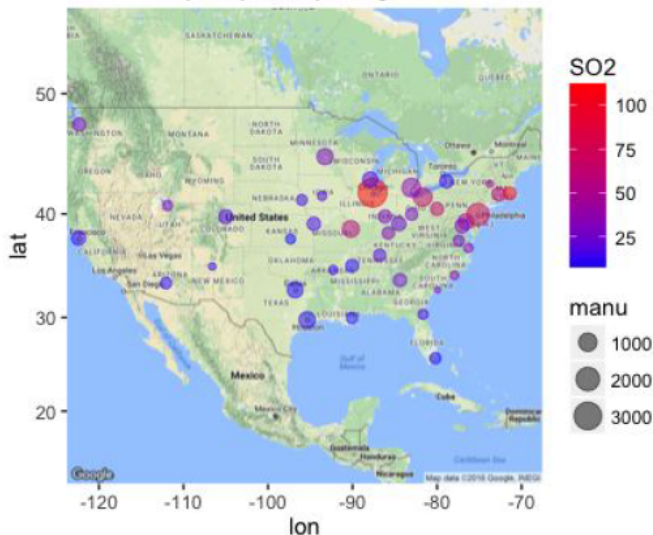


# MAPS

Maps are a very effective communication tool that can visualize thousands of data points in just one figure.



## Map of participating cities

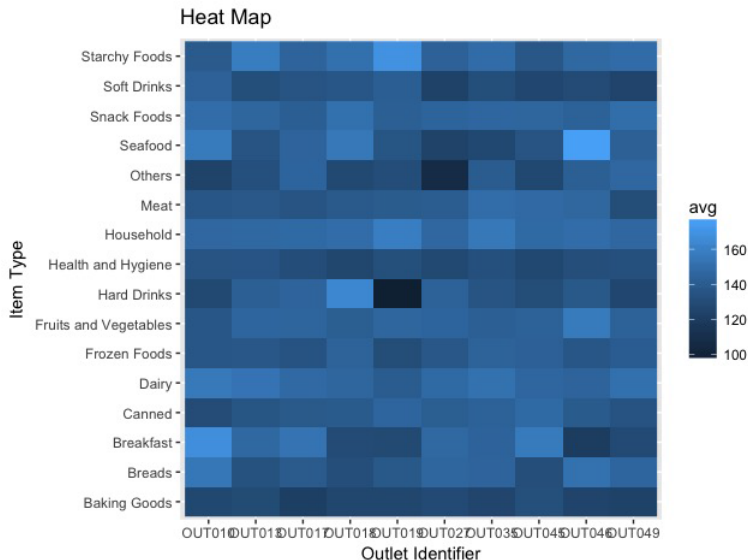




# HEAT MAPS (AND OTHER MATRIX PLOTS)

- A Heat Map uses intensity of colors to display relationship between two, three or many variables in a two dimensional image. It allows you to explore two dimensions as the axis and the third dimension by intensity of color.
- The third dimension could be a measure of similarity or distance between multivariate observations (objects).
- Visit <https://bost.ocks.org/mike/miserables/> for Heat Maps and clustering.

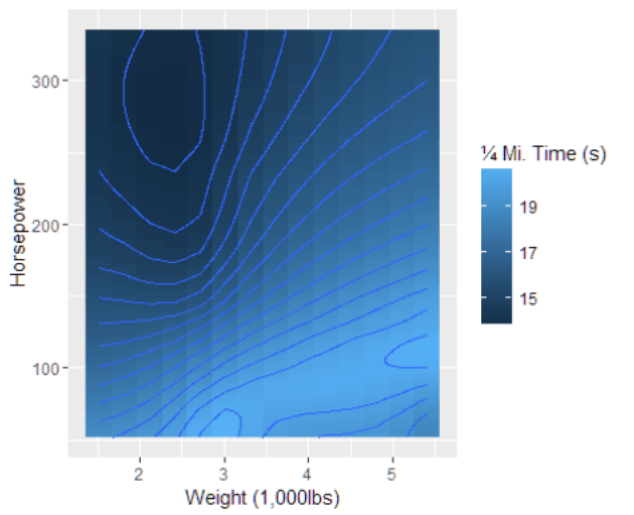
# HEAT MAPS (AND OTHER MATRIX PLOTS)



# CONTOUR PLOTS

- Useful for continuous attributes measured in a spatial grid.
- They partition the plane into regions of similar values.
- The contour lines that form the boundaries of these regions connect points with equal values.
- The most common examples is contour plots of elevation.
- They can also display temperature, rainfall, air pressure, etc.

# CONTOUR PLOTS



# CATEGORICAL DATA - MOSAIC PLOTS

Mosaic plots give a graphical representation of successive decompositions of a data set by Categorical Variables. Counts are represented by rectangles.

Example (*from R. Ihaka's course on Computational Data Analysis and Graphics*) : Who listens to classical music? The table shows a sample of 2300 music listeners classified by age, education and whether they listen to classical music. This is a  $2 \times 2 \times 2$  contingency table.

|              | Education       |           |            |           |
|--------------|-----------------|-----------|------------|-----------|
|              | <i>High</i>     |           | <i>Low</i> |           |
|              | Classical Music |           |            |           |
| Age          | <i>Yes</i>      | <i>No</i> | <i>Yes</i> | <i>No</i> |
| <i>Old</i>   | 210             | 190       | 170        | 730       |
| <i>Young</i> | 194             | 406       | 110        | 290       |

## MOSAIC PLOTS - EXAMPLE

The effect of age and education on musical taste can be investigated by breaking the observations down into more homogeneous groups. The most obvious split is by age. There are 1300 older people and 1000 younger people.

| <i>Old</i> | <i>Young</i> |
|------------|--------------|
| <hr/>      | <hr/>        |
| 56.5%      | 43.5%        |

This is almost certainly a result the way in which the sample was taken.

## MOSAIC PLOTS - EXAMPLE

Within the old and young groups we can now find the proportions falling into each of the high and low education categories:

| <i>Old</i>      |                | <i>Young</i>    |                |
|-----------------|----------------|-----------------|----------------|
| <i>High Ed.</i> | <i>Low Ed.</i> | <i>High Ed.</i> | <i>Low Ed.</i> |
| 30.8%           | 69.2%          | 60.0%           | 40.0%          |

The young group is clearly more highly educated than the old group.

## MOSAIC PLOTS - EXAMPLE

Finally, we can compute the proportion of people who listen to classical music in each of the age/education groups.

| <i>Old</i>      |                | <i>Young</i>    |                |
|-----------------|----------------|-----------------|----------------|
| <i>High Ed.</i> | <i>Low Ed.</i> | <i>High Ed.</i> | <i>Low Ed.</i> |
| 52.5%           | 18.9%          | 32.3%           | 27.5%          |

The music-listening habits of younger people seem to be fairly independent of education level. This is not true for older people.



# MOSAIC PLOTS - EXAMPLE

Summary: From these tables we can see:

- There are slightly more old people than young people in the sampled group.
- The younger people are more highly educated than the older ones.
- The likelihood of listening to classical music depends on both age and education level.

Mosaic plots:

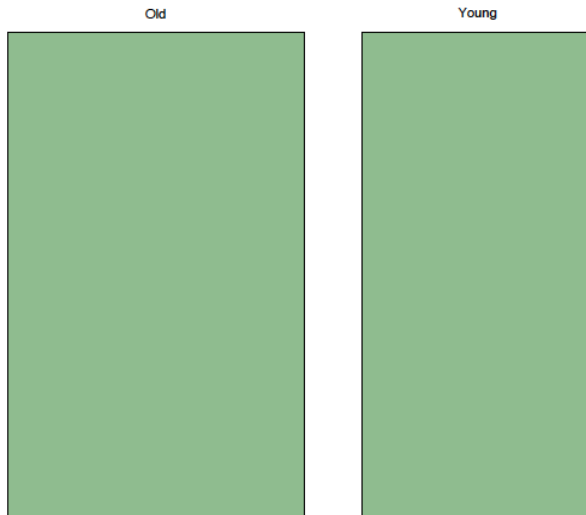
- They give a graphical representation of these successive decompositions.
- Counts are represented by rectangles.
- At each step of plot creation, the rectangles are split parallel to one of the two axes.

# MOSAIC PLOTS - STEPS

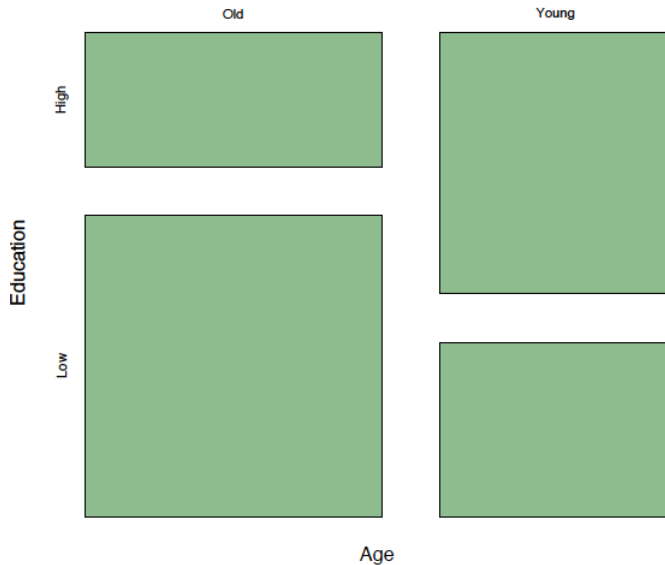
Everyone



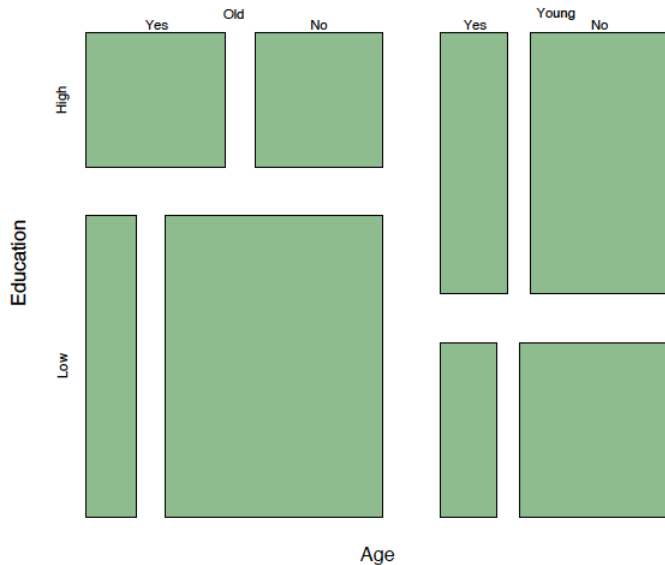
# MOSAIC PLOTS - STEPS



# MOSAIC PLOTS - STEPS



# MOSAIC PLOTS - STEPS



## OTHER EXPLORATORY METHODS

Clustering and anomaly (outlier) detection are usually viewed as exploratory techniques.

So is Principal Component Analysis